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Blockchain meets AI in healthcare: a review of convergent technologies for digital health transformation

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Introduction: Blockchain and AI (Artificial Intelligence) are shaping a new medical ecosystem as two key technologies for the digital transformation of healthcare.

Methods: This paper provides systematic review and theoretical framework for understanding the evolutionary logic, key themes and future trends of blockchain-artificial intelligence integration.

Results and Discussion: This study discusses the digital health transformation path of blockchain and AI in the healthcare field from dimensions such as theme clustering, trend evolution, technical background, application, and future challenges. Blockchain has advantages in building a trusted data infrastructure. Artificial intelligence has a natural talent for enhancing the efficiency of clinical decision-making and operations. The deep integration between the two is driving the transformation of healthcare from single-point digitalization to system intelligence. Despite challenges such as regulatory systems, cross-institutional collaboration, talent structure, and medical ethics, reliable intelligent healthcare systems have become the development direction of future healthcare models.

KEYWORDS

AI (artificial intelligence), blockchain, clinical decision making, digital health transformation, healthcare

1 Introduction

The healthcare industry is at a critical juncture of transformation. Problems such as an aging population, an increasing burden of chronic diseases, rising medical expenses, and the long-term existence of data leakage have made it difficult for the traditional medical system to meet the demands of modern healthcare. Against this backdrop, digitalization and intelligence have become the core trends in the future transformation of healthcare. Blockchain and artificial intelligence are regarded as the key driving forces for building a new medical ecosystem as two disruptive technologies. Blockchain provides a trusted infrastructure for healthcare data governance through distributed ledgers, immutability, traceability, and smart contracts. Artificial intelligence provides intelligent support for improving clinical diagnosis, health management and operational efficiency, such as machine learning, deep learning and natural language processing (Kothinti, 2024).

Blockchain and artificial intelligence are not simply parallel technologies. In modern healthcare, blockchain and AI complement each other. Blockchain can provide high-quality and reliable data for AI, ensuring the accuracy and interpretability of the model. Meanwhile, AI can further optimize the efficiency and resource allocation of blockchain (Hu et al., 2021). However, the integration of these two technologies remains constrained by

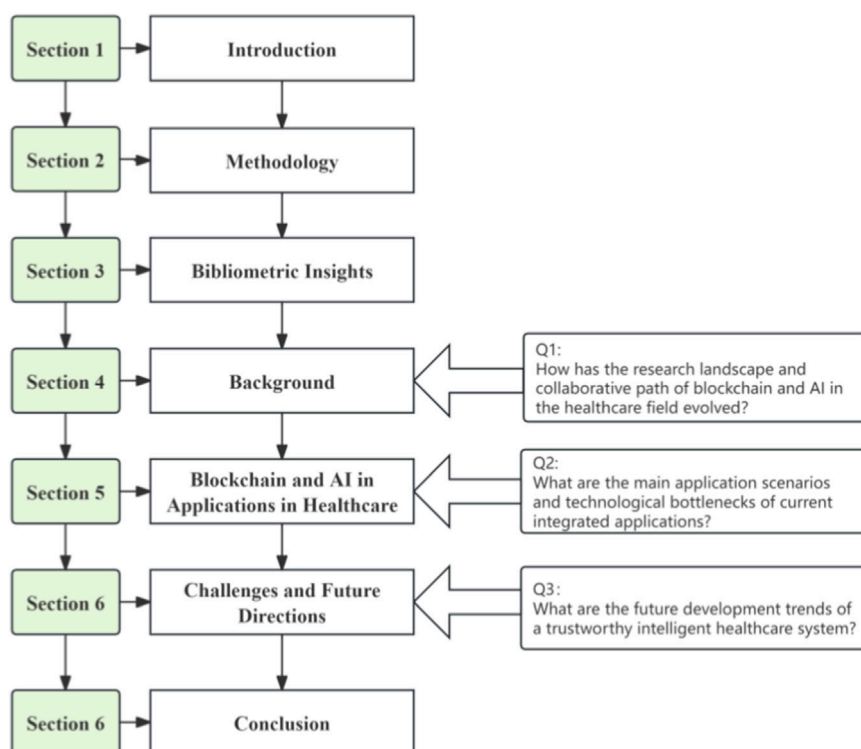


FIGURE 1
The structure of the paper.

numerous technical, regulatory, and organizational barriers. How to leverage these two technologies together to build a trustworthy and intelligent healthcare ecosystem is a more forward-looking perspective (Nikiforova et al., 2024).

With the cross-institutional collaboration needs, the integration of these two technologies is gradually moving from technological exploration to systematic implementation. Although many publications have explored the medical applications of blockchain or AI from a technical perspective, systematic work presenting the integration structure, research hotspots, and trend changes of these two technologies remains limited. Thus, this study, based on a systematic literature search and screening, conducted a literature review and analysis of relevant literature, to answer the following questions.

1. How has the research landscape and collaborative path of blockchain and AI in healthcare evolved? RQ1 is mainly addressed in [Section 4](#).
2. What are the main application scenarios and technological bottlenecks of integrated applications? RQ2 is mainly addressed in [Section 5](#).
3. What are the future development trends of a trustworthy intelligent healthcare system? RQ3 is mainly addressed in [Section 6](#).

This study not only provides quantitative evidence for academia but offers theoretical support for industry practitioners and policymakers to build a forward-looking digital health framework. The article is structured as follows ([Figure 1](#)): [Section](#)

[2](#) introduces the research methods and bibliometric analysis process. [Section 3](#) summarizes research hotspots, knowledge structures and topic evolutions based on bibliometric results. [Section 4](#) systematically reviews the evolution history of blockchain and artificial intelligence in the healthcare field. [Section 5](#) reveals the typical applications of the two technologies in specific medical scenarios. [Section 6](#) presents the current challenges and future development directions. [Section 7](#) discusses the research conclusions and contributions.

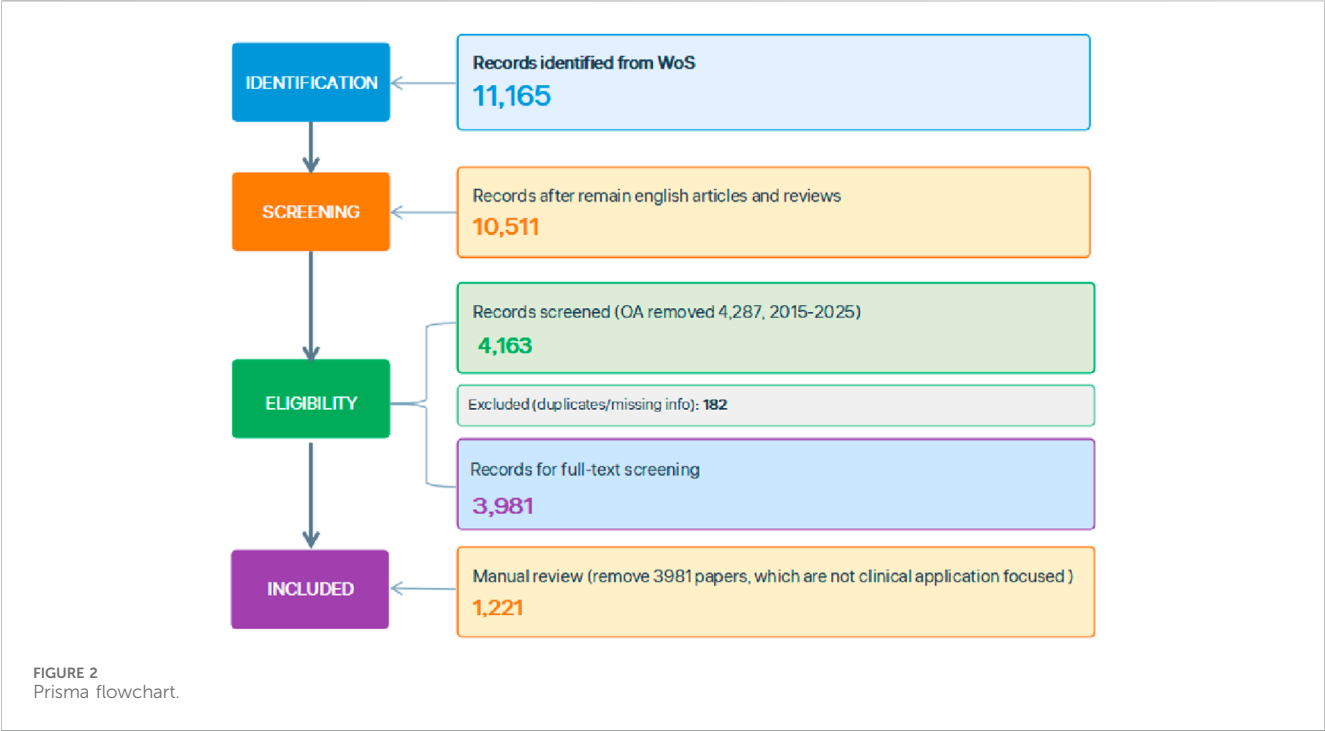
2 Methodology

2.1 Literature screening process

The literature selected for this study was published on the Web of Science (WoS) and IEEE explore from 2015 to 2025. In the WoS database, the search criteria are set to the search subjects of Science Citation Index Expanded (SCI-EXPANDED) and Social Sciences Citation Index (SSCI). The search statement is as follows:

TS=(“Blockchain”and“Healthcare”) OR (“AI”and“Healthcare”) OR (“Blockchain”and“Digital Health Transformation”) OR (“AI”and“Digital Health Transformation”).

The initial search generated 11,165 related articles. Next, it was cleaned up and standardized. First, only the articles written in English and those classified as articles or comments were retained, leaving 10,511. Then, all open access journal articles were deleted, leaving 4,287. To ensure the timeliness of the research, 4,163 articles from 2015 to 2025 were selected.



Finally, in order to avoid overlapping of literature between the Web of Science and IEEE Xplore databases, when merging the search results, this study identified and deleted duplicate literature based on DOI, and also excluded those with missing key information. Each piece of literature was retained only with one record. After processing, a total of 3,981 articles were retained for further analysis.

Based on automatic screening, this study further combines manual review to conduct a second screening of the documents. The retention criteria are literature focusing on the practical application of blockchain or AI in clinical medical scenarios, medical and health data management, the construction of smart medical systems, and the digital transformation of medical and healthcare. After screening, 1,221 documents were finally selected for the descriptive analysis (Figure 2).

We further supplemented the IEEE Xplore database with relevant documents using “AI”, “Healthcare”, “Blockchain”, and “Digital Health Transformation” as keywords, to expand the scope of the study.

Through document coupling analysis, topic clustering analysis, and research trend analysis, the knowledge structure, research hotspots, and development trends of the field in which this paper is located were systematically revealed.

2.2 Descriptive analysis

2.2.1 Main information

The time span of the selected literature’s publication is from 2015 to 2025. It includes 1,221 articles from 451 journals. The average citation count of the selected articles is 30.38. International co-authors account for 41.28%. There are 991 “articles”, the most

TABLE 1 Main information.

Description	Results
Timespan	2015:2025
Sources (journals, books, etc.)	451
Documents	1,221
Average citations per doc	30.38
International co-authorships %	41.28

diverse in category, reflecting the dominant position of innovative research (Table 1).

2.2.2 Annual scientific production

The number of papers has been steadily increasing since 2019. This topic has become a highly popular research frontier from 2019 to 2025 (Figure 3).

2.2.3 Published journals and research directions

According to the statistics in the table “Top 10 Relevant Sources” (Table 2), the journals with the greatest influence and the highest publication volume are mainly concentrated in the fields of computer science, network technology, Internet of Things and medical informatics (Table 2).

Top 10 Relevant Affiliations (Table 3) shows that the research results mainly come from globally leading research institutions with technological advantages. The National Institute of Technology in the United States demonstrated active and sustained contributions, ranking first with 79 articles. Following closely behind are the University of London, Harvard University, King Saud University and the Chinese Academy of Sciences.

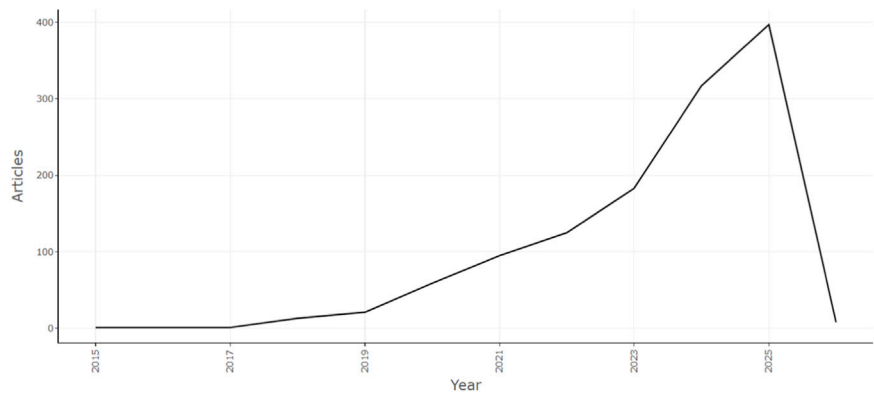


FIGURE 3
Annual scientific production.

TABLE 2 Top 10 relevant sources.

Sources	Articles
IEEE internet of things journal	60
Cluster computing-the journal of networks software tools and applications	40
Journal of supercomputing	23
IEEE transactions on consumer electronics	19
Computers & electrical engineering	18
Peer-to-peer networking and applications	18
Transactions on emerging telecommunications technologies	18
International journal of medical informatics	17
Multimedia tools and applications	16
IEEE journal of biomedical and health informatics	15

2.2.4 Corresponding author’s countries and citation

Table 4 shows the top 10 countries in terms of publication volume and citation volume. The research fields of blockchain, artificial intelligence and healthcare show a distinct multi-center distribution and a pattern of cross-border differentiated influence. India leads with 329 articles. The United States demonstrated the highest number of citations, with a total of 9,917 citations, averaging 49.60 citations per article. South Korea (41 articles) and Australia (27 articles) also have an advantage, with average citation rates of 39.40 times and 49.30 times respectively.

3 Bibliometric insights

3.1 Coupling network analysis

The evolution among blockchain, security, privacy and healthcare applications can be observed from coupled clusters. It emphasizes the establishment of a trustworthy and interoperable

TABLE 3 Top 10 relevant affiliations.

Affiliation	Articles
National institute of technology	79
University of London	39
Harvard university	36
King saud university	36
Chinese academy of sciences	33
Indian institute of technology system	33
Thapar institute of engineering and technology	30
University of Texas system	27
Vellore institute of technology	27
University college London	24

health data ecosystem. The coexistence of each coupled keyword in fields such as blockchain and AI interoperability. It reveals that these two technologies are increasingly regarded as complementary elements (Figure 4).

3.2 Thematic structure of the field

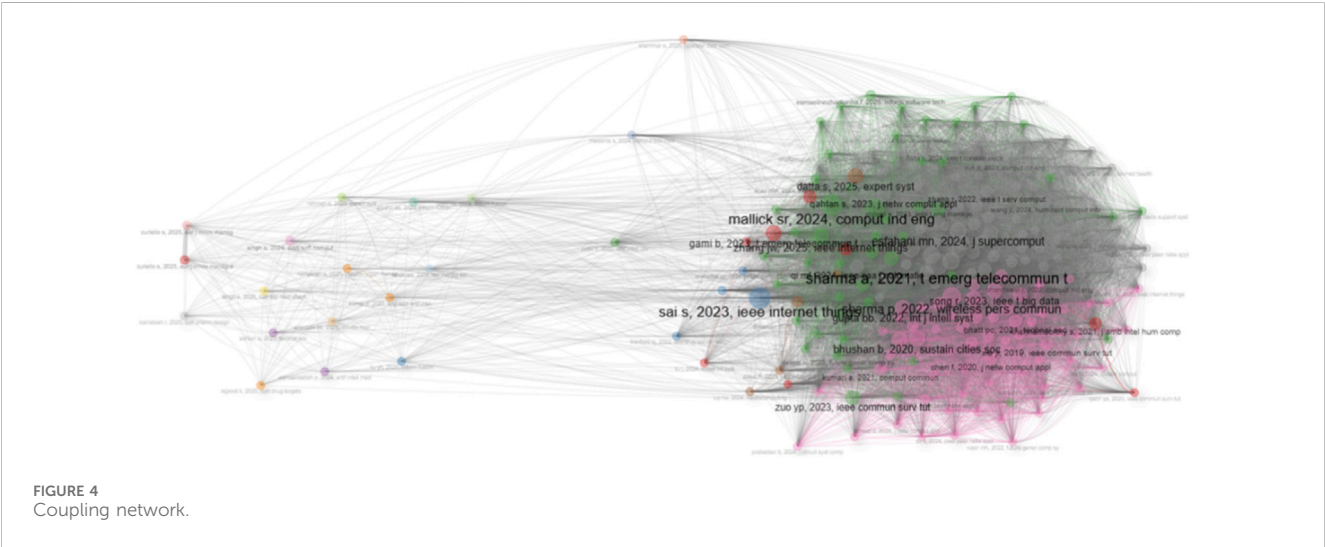
It can be seen from Figure 5 that healthcare, the Internet and topics related to the framework are the most mature. This topic is also very influential because the methods of artificial intelligence and machine learning are deeply integrated into various medical scenarios. The topics of blockchain, security and privacy show highly concentrated and stable research boundaries, especially in trusted data management and privacy protection (Figure 5).

3.3 Trend topics analysis

Research related to blockchain, artificial intelligence and digital health has grown rapidly since 2021. The number of related papers published peaked between 2021 and 2024. The emergence of

TABLE 4 Corresponding Author's countries and citations.

Country	Articles	Total citations	Average article citations
India	329	6,414	19.50
China	219	6,743	30.80
United States	200	9,917	49.60
United Kingdom	42	1,314	31.30
Korea	41	1,616	39.40
Saudi Arabia	33	626	19.00
Australia	27	1,330	49.30
Italy	27	767	28.40
Canada	26	839	32.30
U arab emirates	22	801	36.40



high-frequency keywords such as blockchain, security, privacy, healthcare and artificial intelligence indicates that the field is increasingly focusing on trusted data sharing, privacy-protecting analysis and intelligent medical processes. These trends highlight the rise of blockchain and artificial intelligence and demonstrate that they are the two pillars of the digital healthcare transformation (Figure 6).

4 Background

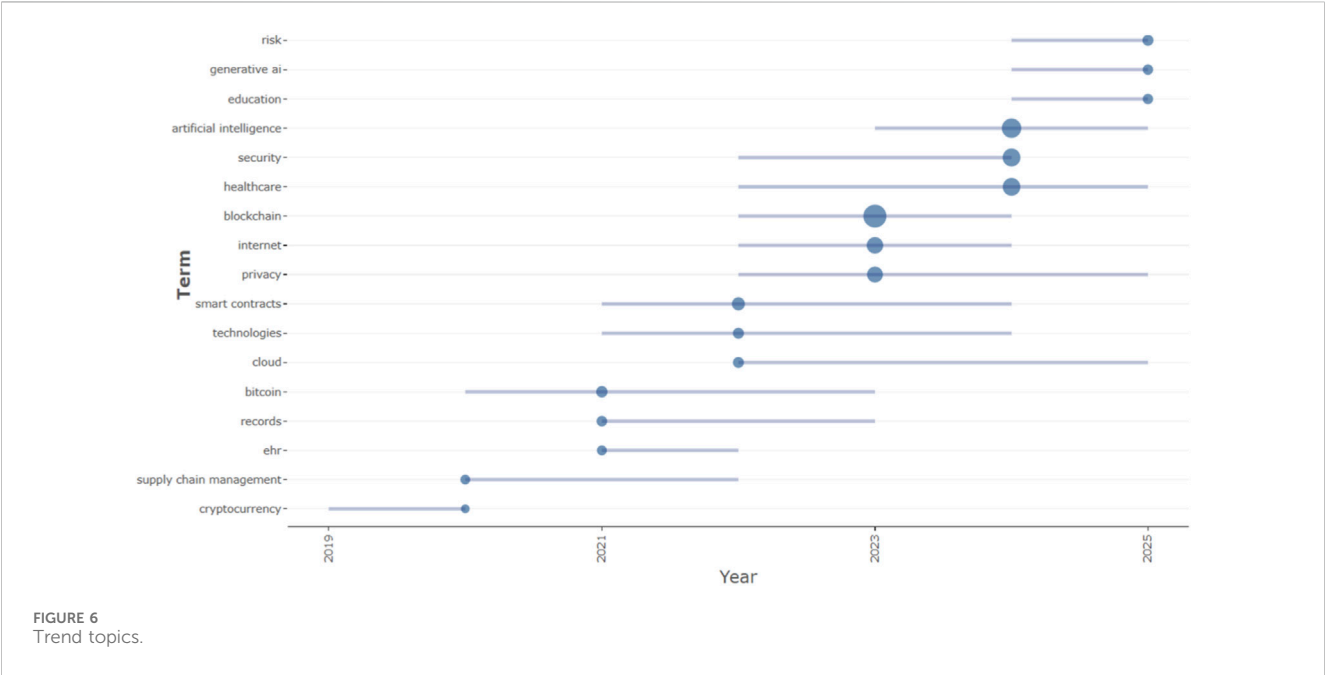
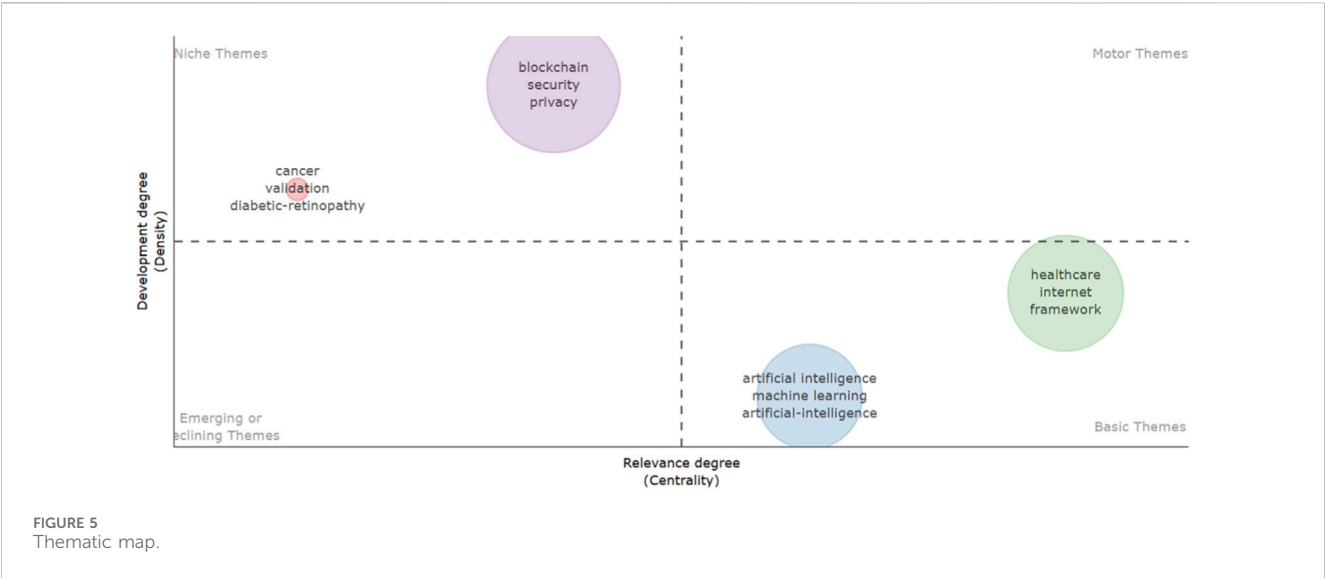
4.1 Features of blockchain and artificial intelligence technologies

4.1.1 Blockchain

Blockchain is a decentralized distributed ledger technology. It includes some features such as decentralization, immutability, traceability, transparency, distributed storage, consensus mechanism, anonymous messaging and smart contracts (Lu, 2019). The core principle of blockchain is to group data into time-linked blocks, each with a unique hash identifier. Each

block contains the hash value of the previous block. It is difficult to hide modifications to blocks because of this structure. If one block's hash value has been modified, then it will change the whole chain. Nodes in the entire network keep maintaining the blockchain via a consensus method. For a new block to be added to the blockchain, it must be verified by most nodes and synchronized to the entire high consistency of data (Xiao et al., 2020).

For information that needs to be preserved and remain unchanged for a long period of time. Blockchain may be suitable option for storing medical information, such as The access logs of electronic medical records (EMR) and clinical trial data (Golec et al., 2023). Decentralization will allow secure sharing of patient data. Immutability ensures the authenticity of medical data. Blockchain ensures traceability in drugs or vaccine is verified based on the traceability of blockchain. Public blockchains address credibility and auditing issues through their complete transparency, with all records being publicly verifiable. Consortium blockchain addresses privacy and business collaboration issues by controlling transparency, with data shared only among authorized entities (Wu et al., 2024). Smart contracts enable blockchains to automatically execute procedural



rules for business processes and reduce human error and fraud (Rahman et al., 2021).

4.1.2 Artificial intelligence

AI is a system of computing technologies capable of simulating, extending, and even enhancing human intelligence. Key AI technologies include multiple branches such as machine learning, deep learning, natural language processing, computer vision, and intelligent robotics (Ali S. et al., 2023). AI technologies learn patterns, rules, and knowledge from massive amounts of data to perform tasks such as classification, prediction, generation, reasoning, and decision-making. Unlike machine learning, which is better suited for analyzing small amounts of data, its subfield deep learning can analyze vast amounts of data. Both technologies are

being applied in medical settings to assist doctors in disease diagnosis and treatment decisions (Bhatia, 2024). Natural language processing extracts structured, machine-understandable information from unstructured text data. Speech processing technology converts human speech into text. Computer vision involves the acquisition and interpretation of information from images and videos and is a core technology in medical image analysis (Deng, 2016).

Federated learning can effectively aggregate data updates from various participants without direct access to their confidential data to protect the privacy of end users (Ali et al., 2022; Lian et al., 2022; Rahmaniar et al., 2025). This has achieved an emerging paradigm of privacy protection. Chatbots and artificial intelligence virtual assistants utilize artificial intelligence technologies and

domain-specific data to perform tasks previously carried out by human experts (Merlo et al., 2023). Robots can be used in healthcare for precise surgical operations (Gupta et al., 2020).

With its powerful self-learning, information processing and predictive capabilities, artificial intelligence can analyze complex, high-dimensional and multi-modal data in medical scenarios. Data such as electronic medical records, medical images and genomic sequences can be transformed into diagnostic recommendations, treatment recommendations and disease progression predictions (Mnasri et al., 2025). AI can help medical staff improve diagnostic efficiency and accuracy and bridge the gap in medical resources between regions. By providing real-time and remote services, it can reduce costs and optimize results (Kumar et al., 2025).

4.2 The development of blockchain and artificial intelligence in healthcare

4.2.1 The development history of blockchain in the healthcare field

Concept Proposal and Introduction Phase (2008–2017). In 2008, blockchain technology began to take shape with the birth of Bitcoin. At that time, blockchain was regarded as an integrated innovation of cryptography, P2P networks, and distributed consensus mechanisms (Zhou et al., 2024). After 2014, blockchain began to be discussed separately from Bitcoin. From 2015 onwards, researchers in the healthcare field began to focus on the decentralized, immutable, and fully traceable characteristics of blockchain, which is potential to solve the problems of trust deficiency and data silos in medical data (Mahmudnia et al., 2022). Against this backdrop, blockchain was proposed for basic functions such as electronic health record (EHR/EMR) management, data sharing, access authorization, and authentication (Khan A. A. et al., 2025). In 2016, the MedRec project released and ran its prototype system, which is one of the early representative cases of using blockchain for electronic health record indexing and access management (Azaria et al., 2016).

Systematic Application Stage (2018–2020). With the surge in medical data and the adoption of inter-institutional medical services, problems such as medical fraud and the circulation of counterfeit drugs have become increasingly serious. Blockchain technology began to move from the research stage to the systematic application stage. Due to the special requirements of medical data regarding privacy and access permissions, consortium blockchain architectures suitable for multi-party collaboration have become the mainstream in the medical field. The U.S. Food and Drug Administration (FDA) require manufacturers to comply with current Good Manufacturing Practices (GMP) when producing pharmaceuticals and medical devices (Livingston and Mattingly II, 2021). Institutions and international pharmaceutical giants have also begun pilot projects for blockchain-based vaccine cold chain management (Mendonça et al., 2021). These applications mark the beginning of blockchain truly being embedded in key processes of the medical industry.

Compliance and Ecosystem Building Stage (2020–Present). After 2020, against the backdrop of global privacy protection, the medical system has placed higher demands on data security, auditing, and compliance. Blockchain, with its traceable, auditable, but not

arbitrarily modifiable characteristics, has become a key technology for building a trustworthy medical data infrastructure. Privacy-preserving computation technologies, such as multi-party secure computation, zero-knowledge proofs, and homomorphic encryption, are beginning to integrate with blockchain (Sun et al., 2021; Zhong et al., 2019). Countries have successively proposed national-level medical data chains or health data space strategies, such as the EU's EHDS, China's Health Big Data Port, and the US ONC's Interconnected Health Ecosystem, expanding blockchain from single-point applications to the digital foundation of the medical field. At this stage, a complete ecosystem of regulation, application, and innovation is being formed, including cross-hospital data sharing platforms, county-level health data chains, end-to-end drug traceability systems, and data trust governance networks combined with AI.

4.2.2 The development history of artificial intelligence in the healthcare field

Early Stages of Intelligent Decision-Making (1970–2000). The exploration of AI in healthcare can be traced back to the 1970s. Expert systems such as MYCIN and DENDRAL, based on rule-based knowledge bases and reasoning mechanisms, assisted diagnosis by encoding doctors' clinical experience (Youheng, 2023). Through expert systems, the medical field embraced the idea of using computational systems to assist clinical decision-making. Humans saw the potential of intelligent technology to improve the quality and efficiency of medical diagnosis. Expert systems have made breakthroughs in logical reasoning, but they also face challenges such as high knowledge acquisition costs. For instance, difficulties in maintaining rule systems or learning from data have led to limited clinical scalability (Ali A. et al., 2023).

Rapid Development Driven by Machine Learning (2000–2015). Although machine learning was applied to healthcare before the 20th century, its widespread application in medicine truly began after 2000. In the 21st century, machine learning has matured (Erickson et al., 2017). The widespread adoption of electronic medical records, the application of medical sensors, and the explosive growth of internet data have spurred the rapid development of machine learning in the medical field (Manivannan, 2024). Researchers began using machine learning algorithms such as statistical learning, support vector machines, and random forests to build models for disease prediction, risk stratification, and medical image recognition. Although their accuracy has not yet reached the level of the deep learning era, this period is to using data rather than manual rules to drive medical intelligence (Rani et al., 2025). Medical institutions also began to accumulate experience in data governance and model deployment, for subsequent technological upgrades.

Deep Learning Drives Intelligent Healthcare (2015–2020). After 2015, the breakthroughs achieved by deep learning in vision, speech, and natural language processing were transmitted to the medical field. CNNs can extract features and patterns from both structured and unstructured data (Ismail et al., 2020). RNNs and LSTMs have shown good performance in patient state prediction and clinical event detection (Bhavya and Pillai, 2019). NLP models have demonstrated unprecedented capabilities in tasks such as automatic medical record summarization, coding, and nursing record analysis. This stage marks the transition of AI from

research to a clinical tool. Furthermore, deep learning has also transformed pharmaceutical research and development, such as protein structure prediction and molecular property prediction, into multiple aspects of the healthcare ecosystem (Ekins, 2016).

The era of Generative Artificial Intelligence and Large Models (2020 to present). With the introduction of Transformer architecture in 2017, deep learning entered the era of pre-training and self-supervised learning. Generative artificial intelligence and large language models have changed the boundaries of medical artificial intelligence capabilities from task-oriented tools to collaborative partners after 2020 (Sun et al., 2025). Models such as the GPT series, Med-PaLM2 and BioGPT can handle multimodal medical data. These large models can support clinical question answering, report generation, auxiliary diagnosis, knowledge reasoning and doctor-patient communication (Luo et al., 2022; Singhal et al., 2025). Generative artificial intelligence, with its superior reasoning and generative capabilities, can effectively reduce the carbon emissions of traditional Internet of Things devices (Prajapat et al., 2025). However, large models also bring new challenges, such as insufficient interpretability, potential biases, privacy risks, and unclear clinical responsibilities (Albahri et al., 2023).

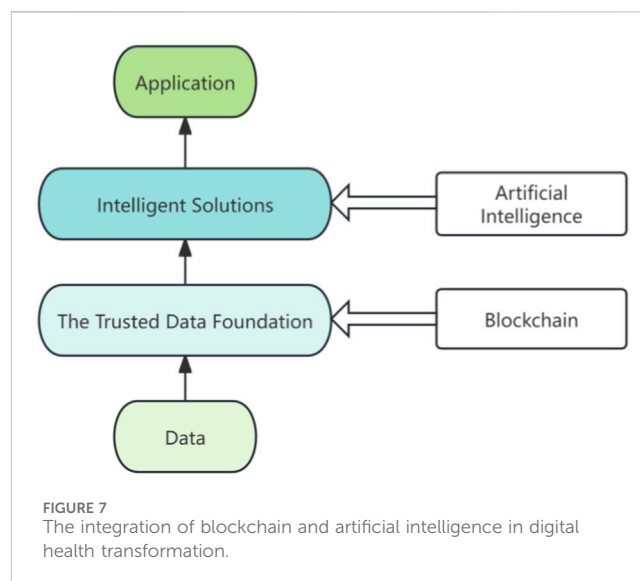
4.3 The integration of blockchain and artificial intelligence in digital health transformation

Blockchain and artificial intelligence are the driving technologies for the continuous transformation of the medical industry. The focus of discussions on blockchain and artificial intelligence in the healthcare sector has shifted from “what they can do” to “how they can become key components of the transformation of the healthcare industry” (Leiva and Castro, 2025). Blockchain and AI together form the foundation of an emerging and reliable smart healthcare ecosystem (Figure 7).

4.3.1 Data and blockchain layer: the trusted data foundation for digital health transformation

Blockchain technology has established a security mechanism for medical data in the digital transformation of health because of the encryption and distributed ledger characteristics of blockchain technology (Esmailnezhad et al., 2025). The immutable storage method and the complete privacy protection function of blockchain effectively control the risk of medical information leakage during transmission and storage. Decentralized identifiers implement verifiable and self-managed digital identity solutions. Patients can precisely control their health data through smart contracts.

In addition, blockchain provides a secure and reliable foundation for cross-institutional medical collaboration and data sharing. The shared data approach of blockchain addresses the limitations of traditional artificial intelligence in constructing personalized medical models due to the lack of local datasets (Das et al., 2025). Blockchain ensures the authenticity and integrity of the data used to train AI from the source. After artificial intelligence mines and analyzes reliable data, it records the output decisions or predictions on the blockchain. For instance,



blockchain is used to track users’ annotation activities in the medical image annotation platform MarkIt. Then these immutable records are used to distribute rewards to annotators to enhance the transparency and auditability of contributions (Sai et al., 2022). This also solves the problems of privacy leakage and difficulty in tracing source information in generative AI (Luo et al., 2024).

4.3.2 AI and application layer: intelligent solutions for digital health transformation

Artificial intelligence (AI) drives more personalized and efficient clinical healthcare by providing precise decision-making and treatment recommendations (Dib, 2025). Intelligent algorithms optimize the allocation of medical resources and improve business processes to enhance the efficiency of healthcare services. Artificial intelligence brings more convenient and higher-quality medical services to patients.

The analysis results of AI can serve as triggers for the execution of blockchain smart contracts, making the system more efficient. AI can also optimize the operational efficiency of blockchain networks and detect malicious behaviors within blockchain networks (Datta et al., 2025). For instance, optimizing a blockchain network involves dynamically allocating resources and predicting network congestion to optimize transaction costs.

Blockchain is the foundation of data security. Artificial intelligence is driving digital health towards higher efficiency and accuracy. The combined application of the two is a patient-centered, data-driven digital health environment. However, both blockchain and AI need to meet the ethical and fairness requirements of the healthcare system (Abdullah et al., 2020).

4.3.3 The trend of integration and development of blockchain and artificial intelligence

The quality of the data used to train artificial intelligence determines the capabilities of the artificial intelligence system. Blockchain records the life cycle of data through its immutable chain structure, including the collection, transmission and use of medical data. This provides AI with high-quality training data and a traceable data foundation (Khan A. et al., 2025). On this basis,

reliable intelligent prediction, diagnosis and treatment plan recommendation by artificial intelligence is particularly important in medical scenarios with extremely low margin of error (El-Saleh et al., 2025).

The efficiency improvement of blockchain application in healthcare can be achieved through artificial intelligence. AI algorithms can optimize the operational efficiency of blockchain, such as dynamic block size adjustment and intelligent partitioning (Ahmed et al., 2023). The data of drug circulation in the blockchain can be analyzed by artificial intelligence. Realize the important function of identifying abnormal transactions and preventing medical insurance fraud while predicting the demand for medical supply (Kapadiya et al., 2025).

The application of AI technology in healthcare faces challenges in privacy protection and regulation due to the high sensitivity of medical data (Li et al., 2023). Privacy protection technologies such as federated learning enable artificial intelligence to train and infer without exposing raw data (Nguyen et al., 2022). Federated learning combined with blockchain technology can ensure compliance and traceability. This ensures the trusted flow of data among institutions and fully realizes its value under strict privacy protection (Myrzashova et al., 2023).

The medical industry is moving from digitalization to a highly unified stage of intelligence and trustworthiness with the integration of blockchain and AI. Blockchain is the foundation of trust and order, while AI provides the ability of wisdom. They jointly construct a truly trustworthy intelligent medical system (Zuo et al., 2023). Utilize blockchain to ensure data security and trusted boundaries, and combine artificial intelligence systems to provide strong support for clinical decision-making, drug development and health management (Firouzi et al., 2022). This model will become the core model of the future healthcare system.

5 Blockchain and ai applications in healthcare

5.1 Blockchain electronic medical records

In the healthcare sector, blockchain technology makes it possible for the doctor and patient to access and share the medical record securely and efficiently. The technology that puts value behind machine codes or some limited resources forms the basis of blockchain. Patients now have ultimate control over their data, changing the model from institutional control to individual control. Blockchain's decentralized identifiers and self-sovereign identity systems empower patients to create a verifiable and controllable digital identity. Through smart contracts, patients can authorize finer access rights to their personal health data. This indicates that medical institutions or Artificial Intelligence models can only make requests or access relevant data with the patient's consent. The risk of interception or misuse of data is minimized through Blockchain technology (Veeramakali et al., 2021).

Blockchain does not directly store massive amounts of original medical records, but records encrypted hash values or indexes of key data. By establishing a unified medical record blockchain, patients can securely own their data. Any

healthcare provider (such as hospitals and pharmacies) can verify the authenticity of patient medical records by comparing the on-chain hash value with the locally encrypted information to confirm the integrity and validity of the data (Mantey et al., 2022). MedRec, proposed by the MIT Media Lab in 2016, was the first blockchain prototype system for access control of health records (EHRs). This system uses blockchain to manage access permissions, giving patients back control of their medical data (Ekblaw, 2017). Patient data can be shared between hospitals. Medicalchain, through public and consortium blockchain architecture, has implemented a full process blockchain system for the healthcare process. This system achieves basic cross-institutional patient data sharing and privacy protection functions and implements a smart contract-supported medical payment system and telemedicine services (Zhou et al., 2024).

5.2 AI medical system

Artificial intelligence (AI) is driving the precision and automation of healthcare services through the deep integration of blockchain and IoT technologies. In the field of remote health monitoring, IoT can monitor patients' physiological signals in real time, such as electrocardiograms, blood glucose levels, and electroencephalograms. Blockchain technology ensures the security of sensitive data through decentralized identity authentication and data encryption. Ultimately, AI algorithms analyze trusted data to achieve remote medical diagnosis and early patient warnings (Aditya et al., 2025; Ahmad et al., 2024).

AI can also enable early disease intervention and personalized health management. The advantage of medical AI models lies in their ability to learn from massive amounts of multimodal patient data to build decision-making models. Training data for the system includes patient blood test data, X-rays of suspicious lesions, and genetic information from biopsy tissue. The AI system will integrate various information to formulate comprehensive diagnostic and treatment plans when new data is input. Researchers have proposed a solution integrating extended Merkel signatures, consortium blockchains, and autonomous AI. The goal of this research is to enhance intelligent diagnostic capabilities while ensuring the security and trustworthiness of electronic medical records (He et al., 2025). Existing research provides many insights for practical applications. Artificial intelligence's predictive capabilities are revolutionizing preventative medicine. Machine learning and deep learning algorithms can analyze population data to identify the potential for certain diseases before patients develop symptoms (Shammi et al., 2024).

The application of artificial intelligence is expanding into broader digital health scenarios. Federated learning-based AI models are being applied in the medical metaverse. This federated learning-based medical metaverse improves performance in several key areas, including diagnosis, monitoring, medical education, and drug development, while ensuring data privacy (Bashir et al., 2023). The synergy between AI and technologies such as blockchain, 5G communication, and drones can enable diversified precision medicine services and autonomous disease management (Wazid et al., 2025).

5.3 Improve the operational efficiency of the healthcare system

AI is improving the operational efficiency of healthcare systems across multiple dimensions, from clinical diagnosis and patient services to administrative management. The core value of healthcare AI is to release healthcare professionals from repetitive work. It automates processes, optimizes allocation of healthcare resources and provides data insight for professionals to concentrate on more valuable diagnostic and treatment work of medical services (Xie et al., 2025). Artificial intelligence enhances the decision-making process by making it faster and more accurate in terms of clinical diagnosis and patient services. AI algorithms can analyze medical images, accurately detecting early signs of diseases such as cancer and even predicting a patient's likelihood of developing the disease beforehand and before any signs appear (Bollineni et al., 2025; Mehrtabar et al., 2025). This creates circumstances for early intervene and there reducing the time needed for diagnosis process. AI virtual health assistants answer patient question, conduct appointments, and follow patients' symptoms and behaviors continuously. It helps patients interact with the doctors easily. Artificial intelligence, when combined with data from wearable devices, can continuously monitor the health status of patients outside of a clinical setting. In addition to that, it can alert the staff whenever there is an anomaly detected. The change is from a passive in-hospital model to an active end-to-end health management (Shammar et al., 2025).

At the administrative and logistical level, AI's automation capabilities reduce operating costs. Technologies like robotic process automation (RPA) have automated scheduling, billing and claim processing. As a result, staff could focus on higher-value tasks such as patient care and strategic planning (Suresh et al., 2024). Robotics and automation technologies have also changed clinical and logistical operations. Robot-assisted surgical systems enable more precise and minimally invasive procedures, to shorten patient recovery time and improve treatment outcomes (Erol et al., 2025). In hospital logistics, AI robots are widely used in warehouse management, drug delivery, and ward cleaning and disinfection. It improves the efficiency of material flow and helps control the risk of hospital-acquired infections.

5.4 Medical supply chain tracking

From production to patient use, pharmaceuticals go through multiple stages, including manufacturing, packaging, distribution, warehousing, transportation, and retail. For example, vaccines and insulin are temperature-sensitive, and the transportation chain for pharmaceuticals is lengthy. If transparency and control are insufficient, problems such as rampant counterfeit drugs and difficulties in auditing by regulatory agencies can arise. Blockchain, with its characteristics, has become a suitable underlying technology for tracking the pharmaceutical and healthcare supply chain (Pradhan et al., 2023).

In practice, blockchain is integrated with technologies such as the Internet of Things (IoT) and smart contracts to achieve functions such as anti-counterfeiting and source verification of pharmaceuticals, cold chain tracking of items like vaccines,

management of medical devices and reagents, automated management of drug recalls, and cross-border drug tracking and distribution (Shinde et al., 2024). By recording information from each stage of pharmaceutical production and distribution on the blockchain, an immutable circulation log is formed. Pharmacies and other terminals can easily review and verify whether the source warehouse and transportation history of pharmaceuticals meet standards and authorize patients to use them after confirmation. For example, FarmaTrusts is a blockchain-based global drug tracking system where regulatory agencies can view on-chain information in real time if drugs are not repackaged (Rojnic, 2022). Blockchain technology is a core technology for global pharmaceutical supply chain compliance and risk governance, such as combating counterfeit drugs, improving public health security, and building a smart healthcare supply chain with cross-institutional collaboration.

Artificial intelligence (AI) technology is shaping smart supply chains. AI can predict the sustainability of healthcare supply chains based on data. By optimizing resource allocation, operational efficiency, and environmental and social impacts, AI can help healthcare supply chains achieve more efficient, economical, and sustainable operations (Azadi et al., 2023).

5.5 Personalized medicine

Using AI and robotics is a long-term scheme in the healthcare sector. The use of these technologies creates a place for them in the existing healthcare ecosystem and existing workflows making healthcare affordable and patient-centric (Herrero Antón de Vez et al., 2025).

In the field of medicine, AI is changing the way doctors diagnose diseases. Artificial Intelligence can analyze massive amounts of patient data. This lets us identify patterns and help doctors and physicians in making decisions. AI technology is being utilized to hasten drug development. AI can identify candidates and predict drug efficacy. This can result in more targeted and effective treatments. The European Union and other global players are creating standards, frameworks and related measures for personalized precision medicine (ICPerMed) (Causio et al., 2022).

Mobile health applications have made AI more accessible and acceptable too, on the patient side. Patients can monitor their health using an AI health app that also gives personalized advice (Sakthi et al., 2024). Certain smart wearable devices such as Fitbit and Apple Watch use IoT and AI technologies to constantly monitor the health and activity data of the users. This data is synced to a mobile app for analysis. This device has user-friendly charts that simplify the monitoring of body data and personal health.

6 Challenges and future directions

As healthcare organizations incorporate blockchain and artificial intelligence technologies into their internal systems, they face challenges that go beyond AI algorithm performance issues, data processing and systems deployment challenges. These challenges and future research directions include institution



building, resource allocation, inter/transdisciplinary collaborations, and ethical constraints (Figure 8).

6.1 Regulatory systems adapted to new technologies

The current regulations governing the use of AI in healthcare are targeted at physical medical devices or static database systems. However, the systems of AI can learn and grow forever. When AI systems are deployed in the dynamic environment of healthcare, one could rely on regulation and legislation frameworks, which are based on the static nature and repeatability of the software (Omidian, 2024), limit the learning and improving possibilities of an AI algorithm. In inability to get updated data, models can make erroneous diagnosis. This can happen when they are not trained enough.

Current regulatory frameworks for AI are inadequate since they need improved legal mechanisms and frameworks. The first step would be to create more transparent reporting. The current research suggests having more transparent reports authoring mechanisms to report the issue being faced. AI developers achieve interpretability through explanations of how AI models learn and evolve and how they are being supervised (Babic et al., 2021). This is to ensure at every step that the training data is fair and there is no data bias in conclusion, it is imperative for AI developers, health practitioners and policy economists to work together to formulate a new regulatory framework that ensures compatibility.

6.2 Professional talent team and robotics team

The future of healthcare teams is challenged by talent transformation and technology. As it pertains to producing a cohort of human resources that possess the ability to deal with the future trends of healthcare, many healthcare professionals do not have sufficient insights on blockchain and AI and fear its consequences and impacts (Pee et al., 2019). Some even feel like AI will expose their private life or take away their jobs. AI can help to augment the competence of human health professionals, not replace them. Human judgment and empathy cannot be replaced by machines (Sezgin, 2023). Training and empowering staff to learn constantly and collaborate with machines are essential for transforming talent (Rajmohan et al., 2024).

The aging population is causing a shortage of doctors, nurses, and other health professionals to equip technology with. Healthcare systems are mostly overwhelmed during epidemics of large scale (Ogbuke et al., 2023). Healthcare professionals can be trained in AI skills to enable deployment of AI-based healthcare solutions. In addition, it can replace humans in doing basic and mechanical jobs to lessen the burden of the health workers (Kyrarini et al., 2021).

6.3 Integrating multiple technologies to build an intelligent medical platform

To solve the communication issues between patient-doctors and unorganized health management and the various inconveniences caused within the healthcare system which comes with asymmetric information, to build a one-stop intelligent platform that integrates multi-source technology and medical resources is of great practical significance. The structured integration of key data like electronic health records, treatment pathways, medical costs can reduce operational barriers and repeated manual work in the patient's medical process, based on a patient-centered design philosophy (Joda et al., 2019).

At the service level, the diagnostic process can achieve intelligent triage and personalized treatment planning based on a series of clinical guidelines. Combining deep learning with blockchain helps prevent adversarial attacks and protect privacy (Lim et al., 2022; Rajmohan et al., 2024). Therefore, a precise doctor-patient collaborative decision support system based on natural language processing, machine learning, and deep learning algorithms needs to be built at the technical level (Gami et al., 2023; Messinis et al., 2024). Data transparency and enhanced user interfaces have the potential to fundamentally change the doctor-patient trust relationship. Cloud computing and blockchain architectures still face network congestion and high latency issues when processing massive amounts of data (Khanh et al., 2024). Blockchain-based edge fog computing integration is being used to build collaborative architectures for intelligent healthcare systems (Rathore et al., 2025). The next-generation communication technology 6G will ensure the upgrade of edge computing in the healthcare field and promote the development of federated learning and generative artificial intelligence models through data throughput, mobile coverage, and security (Taleb et al., 2025). As the core infrastructure for future smart healthcare, 6G will provide decisive support for AI-based medical care and real-time remote diagnosis and treatment (Ahad et al., 2024).

6.4 Ethical issues of ensuring security and privacy

The equitable application of artificial intelligence (AI) in healthcare is a challenge. The "black box" nature of AI leads to a lack of patient trust. Current issues include data privacy, algorithmic bias, and equitable access to healthcare services. Patient and prescription data are highly sensitive and require rigorous classification and protection. Furthermore, the datasets used to train AI algorithms do not represent the diversity of the entire population. Subjective biases from algorithm engineers are inevitably incorporated into algorithm design. Therefore, AI algorithms may provide biased treatment recommendations to women or other specific groups (Lian et al., 2023). Blockchain technology offers a secure solution due to its immutability and

decentralization (Ren et al., 2022). Fairness and reduced bias should be ensured throughout the entire lifecycle of AI algorithm development and iteration.

Technology cannot solve all ethical problems. Finding a balance between ensuring data security and guaranteeing urgent access presents a new challenge. However, technology cannot solve all ethical problems. For instance, to enhance operational efficiency, AI faces challenges. The management costs, quality of services, efficiency of the health supply chain and the exchange of information between different healthcare systems are just some of the challenges (Omidian, 2024).

7 Conclusion

This paper developed from large scale bibliometrics, and reviewed analysis reveals the evolution and application path of the integration of blockchain and AI in healthcare. According to the research, the building blocks of AI and blockchain can be ingrained in any healthcare solution that needs trustworthy and auditable digital systems for health data. It is the trend of the present to integrate the two for digital health transformation.

With trusted data and intelligent analysis, application-level solutions such as electronic medical record management, intelligent diagnostic systems, virtual assistants, supply chain tracking, personalized care, etc. are improving quality of services and better use of resources. Implementation of integration is still being affected by several factors, notably and among those regulatory adaptation, privacy protection and so on. The future of healthcare calls for a dynamically adaptive regulatory framework and intelligent system that integrate AI and blockchain, to create a fair and transparent ecology of digital health transformation.

Data availability statement

The original contributions presented in the study are included in the article/supplementary material, further inquiries can be directed to the corresponding author.

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